

AI/ML Project links

1. LLM-powered RAG (Retrieval-Augmented Generation)

Why it matters: This is the #1 project for AI in 2025. It moves you from just "using" ChatGPT to building a system that can chat with specific, private documents (PDFs, internal wikis) without retraining the model.

- **Key Concepts:** Vector Databases (Pinecone/Chroma), LangChain, Embedding models.
- **GitHub Link:** [Microsoft: Generative AI for Beginners](#)
- **Why sooner?** Understanding how to ground an AI in specific facts is more valuable today than building a custom LSTM from scratch.

2. End-to-End MLOps Pipeline

Why it matters: In the industry, "the model" is only 5% of the code. The rest is plumbing. This project teaches you how to automate training, track experiments, and deploy a model as a web service.

- **Key Concepts:** MLflow (experiment tracking), DVC (data versioning), GitHub Actions (CI/CD), FastAPI.
- **GitHub Link:** [PatrickCmd/mlops-project](#) (A comprehensive example using Prefect, MLflow, and AWS).
- **Why sooner?** It stops you from losing track of which version of your code produced which result.

3. Real-Time Object Detection (YOLOv8/v10)

Why it matters: Computer vision isn't just about classifying static images. Real-world value lies in real-time video processing—counting people in a store, detecting safety gear on a construction site, or tracking sports players.

- **Key Concepts:** YOLO (You Only Look Once), OpenCV, Model quantization (making it run fast on low-power devices).
- **GitHub Link:** [Ultralytics YOLOv8](#) (The industry gold standard for CV implementations).
- **Why sooner?** It teaches you the performance constraints of AI—how to balance accuracy vs. speed.

4. Recommender Systems (Personalization)

Why it matters: This is the engine behind Netflix, Amazon, and TikTok. It's one of the few ML projects where "Accuracy" isn't the only metric—you also have to care about "Novelty" and "Diversity."

- **Key Concepts:** Collaborative Filtering, Matrix Factorization, Deep Recommenders (Two-Tower models).
- **GitHub Link:** [Microsoft Recommenders](#) (The most complete library of industry-standard recommendation algorithms).

- **Why sooner?** It shifts your mindset from "Predicting a Label" to "Understanding User Behavior."

5. Fraud & Anomaly Detection (Imbalanced Data)

Why it matters: Most tutorials use balanced datasets (50% dogs, 50% cats). In the real world, fraud is 0.01% of transactions. If your model just says "Not Fraud" every time, it's 99.9% accurate but 100% useless.

- **Key Concepts:** SMOTE (oversampling), Isolation Forests, Precision-Recall curves (over Accuracy).
- **GitHub Link:** [Fraud Detection Loan Project](#) (Search for the Fraud section in this massive industry repo).
- **Why sooner?** It forces you to learn that "Accuracy" is often a lie in real-world data science.